

AWS Certified Developer Associate Training

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TGS-2025052675 | Tertiary Infotech Academy Pte Ltd | UEN 201200696W

Trainer: Dr. Alfred Ang | Version 2

Digital Attendance (TRAQOM - SSG)

 Scan the TRAQOM QR code on the LMS at required checkpoints.

 Use the identity registered for this course.

 Attendance is mandatory and recorded digitally.

About the Trainer



Your Trainer

General Trainer template -
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS

PhD - specialises in AI, automation, software engineering and cloud learning.

DELIVERS

WSQ courses on cloud fundamentals, AI, automation and app development.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Learner Introduction



Name and role



AWS development experience



One application you want to build



Your goal for this course

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively - no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

Be punctual; return from breaks on time.

6

75% attendance is required.

Course Outline

Day 1

Labs 1-2

Secure setup and serverless APIs

Day 1

Labs 3-4

Data and event integration

Day 2

Labs 5-6

CI/CD and deployment

Day 2

Labs 7-8

Security, operations and review

Learning Outcomes

Secure

**IAM +
credentials**

Use least privilege

Build

**Serverless +
data**

Create application patterns

Deliver

CI/CD + IaC

Automate deployments

Operate

**Observe +
optimize**

Troubleshoot applications

Briefing for Assessment

 Open-book assessment on Day 2 from 4:00 PM to 6:00 PM.

 Written Assessment: 1 hour.

 Practical Performance: 1 hour.

 Follow trainer and LMS submission instructions.

Assessment and Funding

 Complete both WA and PP components.

 Meet attendance and assessment requirements.

 Submit answers on the LMS.

 Sign the Assessment Summary Record.

Assessment Flow Reminder



The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

Access the Hands-On Labs

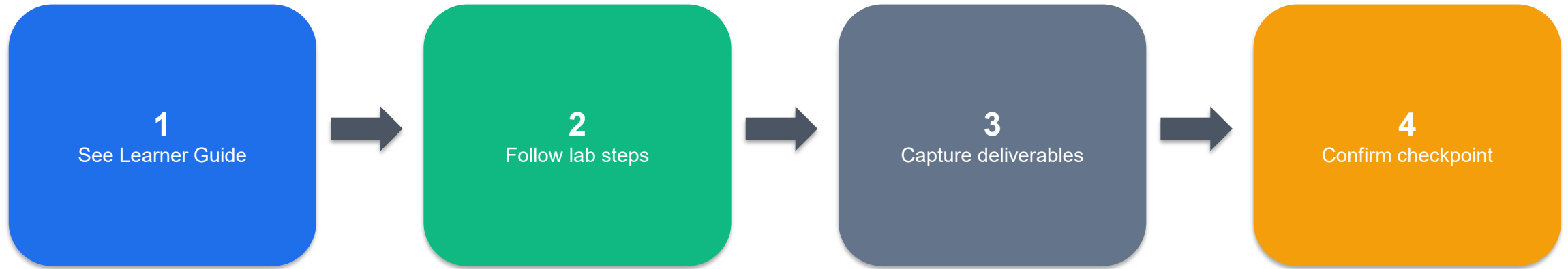
 Option A: git clone <https://github.com/tertiarycourses/TGS-2025052675---AWS-Certified-Developer-Associate-Training.git>

 Option B: GitHub Code -> Download ZIP

 <https://github.com/tertiarycourses/TGS-2025052675---AWS-Certified-Developer-Associate-Training/tree/main/labs>

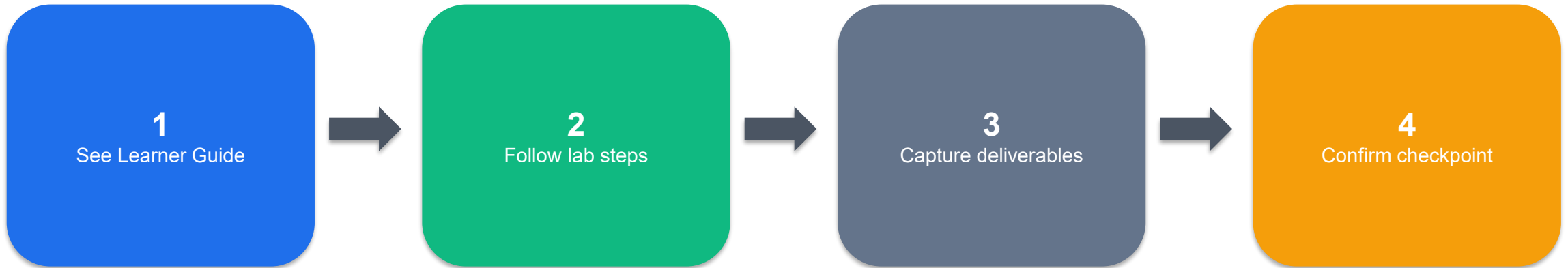
 Use the AWS account and permissions assigned by your trainer.

AWS Developer Learning Flow



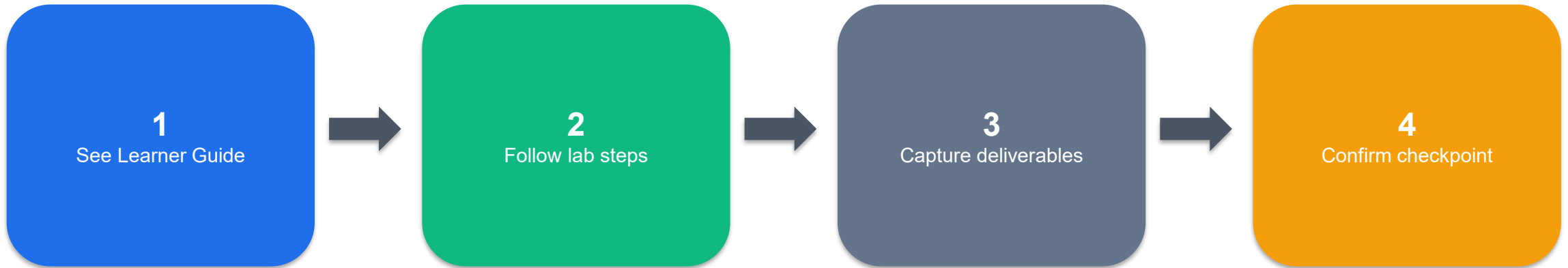
All artifacts follow the same authoritative lab sequence.

Services Practised



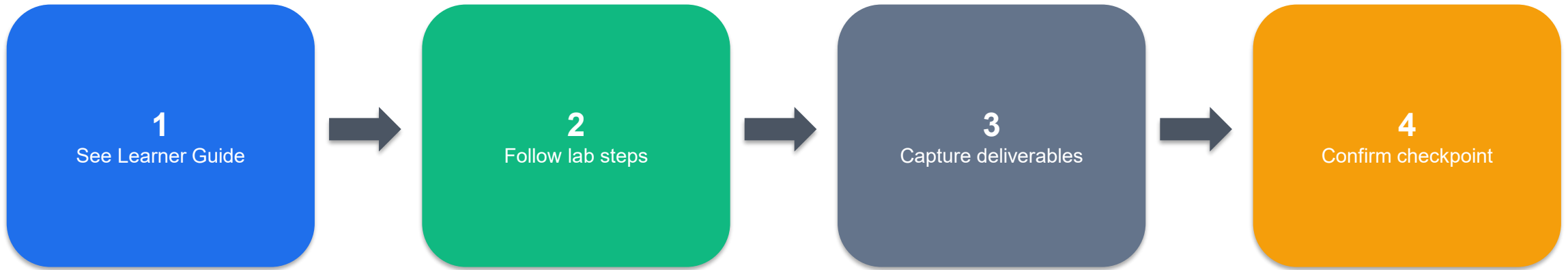
All artifacts follow the same authoritative lab sequence.

Secure Delivery Lifecycle



All artifacts follow the same authoritative lab sequence.

Troubleshooting and Optimization Loop



All artifacts follow the same authoritative lab sequence.

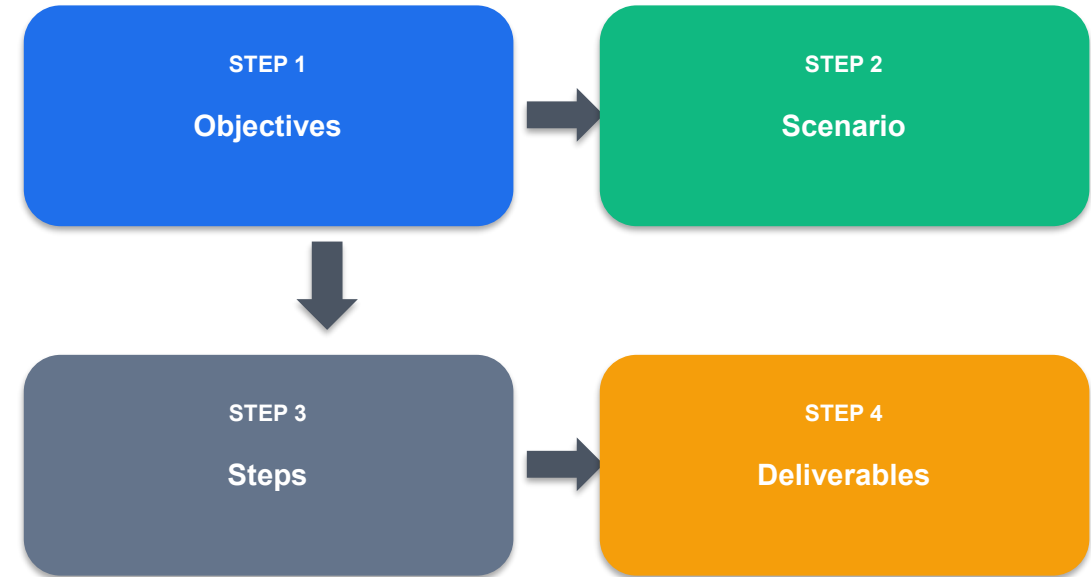
Lab 1 - Account Safety, IAM, AWS CLI, SDKs, and Developer Setup

Domain 1 - Development with AWS Services

Day 1

labs/lab-01-account-
iam-cli-sdk-
developer-setup.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 1 - Objectives

-  Configure a safe AWS developer baseline.
-  Review IAM users, roles, policies, and least privilege.
-  Use AWS CloudShell or AWS CLI.
-  Understand SDK configuration and credential safety.

Lab 1 - Steps 1-5



Sign in to the AWS account assigned by your trainer.



Open IAM.



Review users, groups, roles, and policies.



Confirm you are in the correct region.



Verify MFA is enabled where permitted.

Lab 1 - Steps 6-10



Open an AWS managed policy and identify actions, resources, and effects.



Open AWS CloudShell.



Run ``aws configure list``.



Create or review a developer role only if your trainer permits it.



Run ``aws sts get-caller-identity``.

Lab 1 - Steps 11-15



Review the difference between access keys, temporary credentials, roles, and instance profiles.



Record how the SDK credential provider chain works at a high level.



Save your notes.



Open AWS SDK documentation for one language such as Python, JavaScript, Java, or .NET.



Create a short least-privilege checklist for application development.

Lab 1 - Verify and Clean Up

Verification

1. IAM concept notes
2. CloudShell command output notes
3. Credential safety checklist
4. SDK configuration notes

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep rg-az900-labs only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 16 performs the final course cleanup.

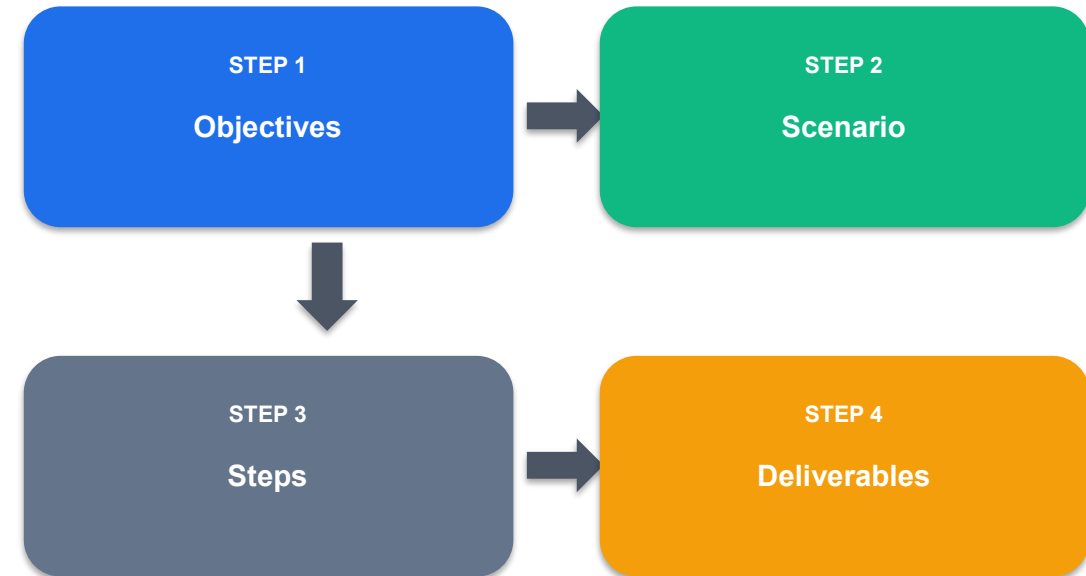
Lab 2 - Lambda, API Gateway, and Serverless APIs

Domain 1 - Development with AWS Services

Day 1

labs/lab-02-lambda-
api-gateway-
serverless-api.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 2 - Objectives

-  Understand Lambda function configuration.
-  Create or review a Lambda test event.
-  Connect Lambda to API Gateway conceptually or hands-on if permitted.
-  Review permissions, logs, timeout, memory, and versions.

Lab 2 - Steps 1-5



Open AWS Lambda.



Choose an approved runtime such as Python or Node.js.



Create a test event.



Create a function only if your trainer permits it.



Review handler, runtime, timeout, memory, environment variables, and execution role.

Lab 2 - Steps 6-10



Run the test event and review the response.



Add or review a simple log statement.



Review concurrency and throttling concepts.



Open CloudWatch Logs for the function.



Review function versions and aliases.

Lab 2 - Steps 11-15



Open API Gateway.



Create a simple route to Lambda only if permitted.



Review Lambda resource-based permissions for API Gateway invocation.



Review REST API, HTTP API, routes, stages, integrations, and authorizers.



Test the route or document the request/response flow.

Lab 2 - Steps 16-17

 Delete any test API and function if instructed.

 Save your notes.

Lab 2 - Verify and Clean Up

Verification

1. Lambda configuration notes
2. Test event notes
3. API Gateway request flow notes
4. CloudWatch log notes

Cleanup Discipline

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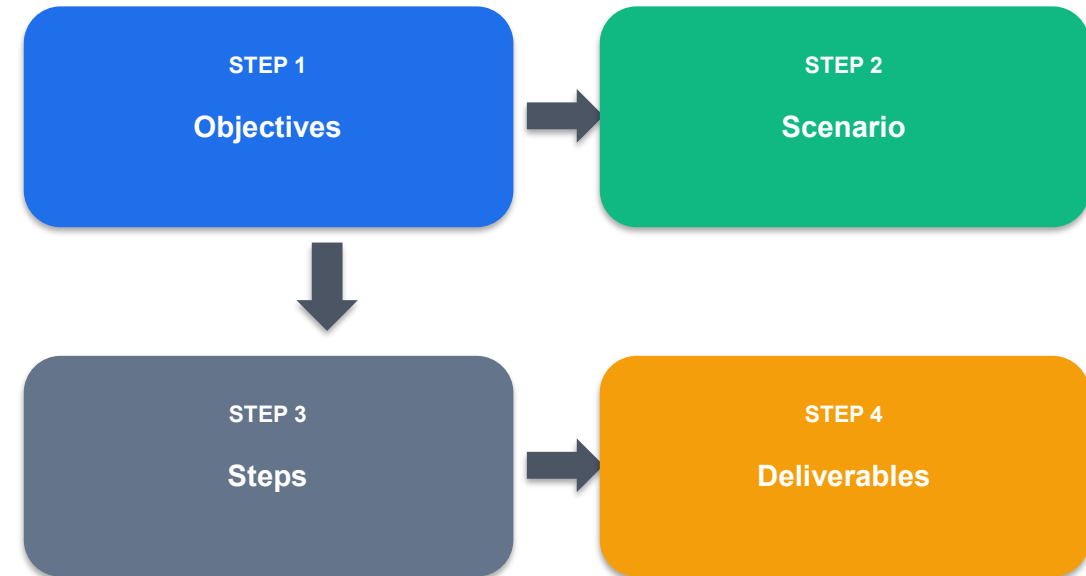
Lab 3 - DynamoDB, S3, and Application Data Patterns

Domain 1 - Development with AWS Services

Day 1

labs/lab-03-
dynamodb-s3-
application-data.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 3 - Objectives

 Model application data using DynamoDB access patterns.

 Review keys, indexes, capacity, TTL, and streams.

 Understand S3 object access and presigned URLs.

 Compare data storage choices for applications.

Lab 3 - Steps 1-5



Open DynamoDB.



Create a table only if your trainer permits it.



Add a sample item if permitted.



Review table, partition key, sort key, item, and attribute concepts.



Choose a sample access pattern such as `Get order by order\_id`.

Lab 3 - Steps 6-10



Review global secondary index and local secondary index concepts.



Review TTL and DynamoDB Streams use cases.



Review bucket, object, prefix, metadata, and versioning concepts.



Review capacity modes and throttling behavior.



Open S3.

Lab 3 - Steps 11-15



Create a bucket only if permitted.



Upload a small sample file if permitted.



Create a table comparing DynamoDB, S3, RDS, and ElastiCache for application data.



Keep Block Public Access enabled.



Review presigned URL use cases.

Lab 3 - Steps 16-17

 Delete temporary resources if instructed.

 Save your notes.

Lab 3 - Verify and Clean Up

Verification

1. DynamoDB access pattern notes
2. Key and index design notes
3. S3 object access notes
4. Application data service comparison table

Cleanup Discipline

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2. Keep rg-az900-labs only when the next lab needs it.
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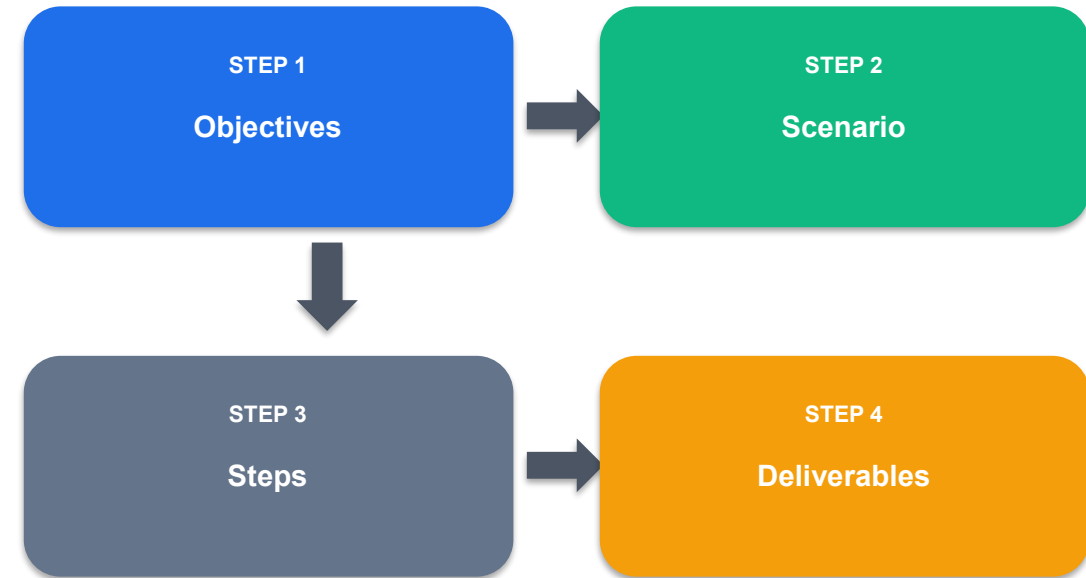
Lab 4 - SQS, SNS, EventBridge, and Step Functions

Domain 2 - Security and Deployment

Day 1

labs/lab-04-sqs-sns-
eventbridge-step-
functions.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 4 - Objectives

 Compare asynchronous messaging and event services.

 Understand queues, topics, event buses, and workflows.

 Design retry and dead-letter queue behavior.

 Map event-driven services to application scenarios.

Lab 4 - Steps 1-5



Open Amazon SQS.



Review visibility timeout, message retention, long polling, and dead-letter queues.



Review topics, subscriptions, fanout, and filtering.



Review standard and FIFO queue concepts.



Open Amazon SNS.

Lab 4 - Steps 6-10



Open Amazon EventBridge.



Open AWS Step Functions.



Create a comparison table for SQS, SNS, EventBridge, and Step Functions.



Review event buses, rules, schedules, and targets.



Review state machines, tasks, choices, retries, catches, and executions.

Lab 4 - Steps 11-15



Design a workflow where an API request queues work for a background processor.



Design a fanout notification pattern.



Save your notes.



Add retry and DLQ handling.



Design a scheduled event pattern.

Lab 4 - Verify and Clean Up

Verification

1. Event service comparison table
2. Queue-based processing design
3. Fanout design
4. Retry and DLQ notes

Cleanup Discipline

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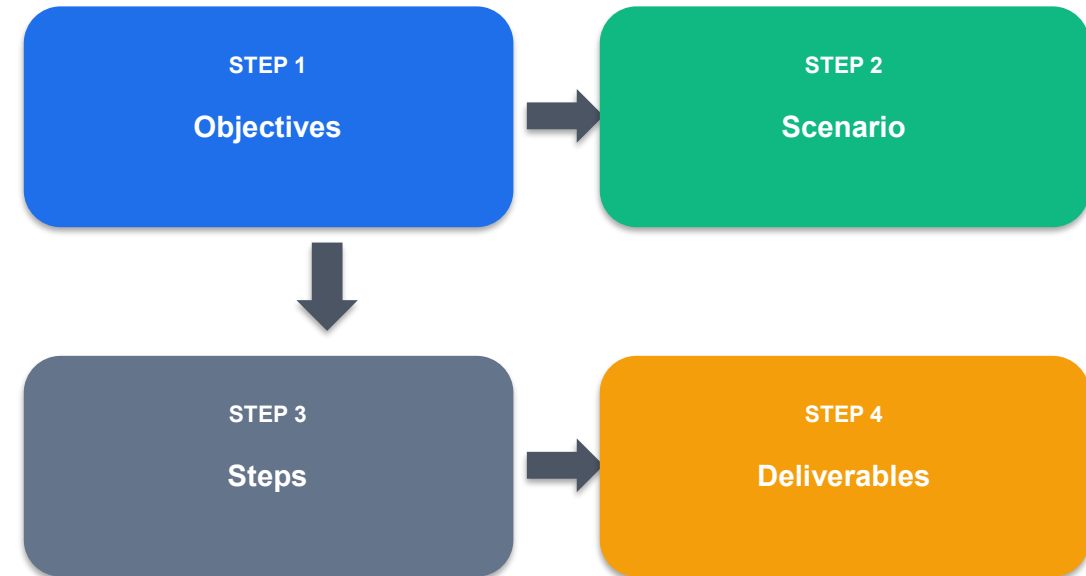
Lab 5 - Developer Tools and CI/CD Pipelines

Domain 2 - Security and Deployment

Day 2

labs/lab-05-
codecommit-
codebuild-
codedeploy-
codepipeline.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 5 - Objectives

-  Understand AWS developer tools for CI/CD.
-  Review source, build, test, deploy, and release stages.
-  Compare CodeCommit, CodeBuild, CodeDeploy, and CodePipeline.
-  Design a simple application delivery pipeline.

Lab 5 - Steps 1-5



Open CodePipeline.



Open CodeBuild.



Open CodeDeploy.



Review pipeline stages and actions.



Review build projects, buildspec files, environment images, artifacts, and logs.

Lab 5 - Steps 6-10



Review deployment groups, deployment configurations, lifecycle hooks, and rollback concepts.



Create a comparison table for source, build, deploy, and pipeline services.



Identify where artifacts are stored.



Open CodeCommit if available or use a trainer-approved external Git provider.



Design a pipeline with source, build, test, package, deploy, and approval stages.

Lab 5 - Steps 11-14

-  Identify where environment variables and secrets should be handled.
-  Compare blue/green, rolling, canary, and all-at-once deployment concepts.
-  Review failure and rollback paths.
-  Save your pipeline design.

Lab 5 - Verify and Clean Up

Verification

1. CI/CD service comparison table
2. Pipeline design
3. Deployment strategy notes
4. Rollback and artifact notes

Cleanup Discipline

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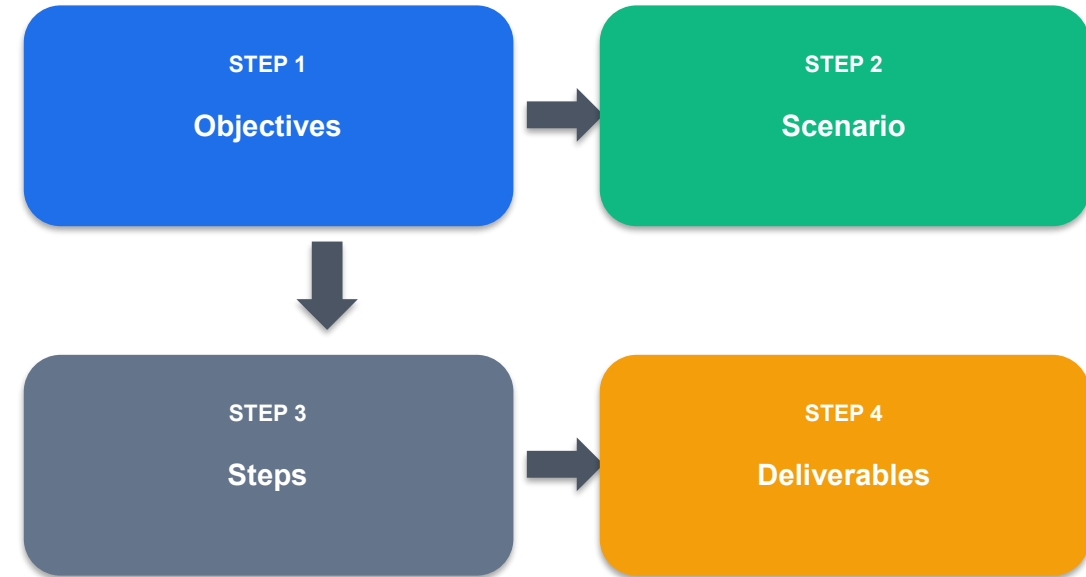
Lab 6 - Deployment with SAM, CloudFormation, Elastic Beanstalk, and Containers

Domain 2 - Security and Deployment

Day 2

labs/lab-06-
deployment-sam-
cloudformation-
containers.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 6 - Objectives

 Compare AWS deployment and infrastructure-as-code options.

 Understand CloudFormation templates and stacks.

 Review AWS SAM for serverless deployment.

 Compare Elastic Beanstalk and container deployment patterns.

Lab 6 - Steps 1-5



Open CloudFormation.



Open AWS SAM documentation or service pages.



Review SAM build, package, deploy, and local testing concepts.



Review stacks, templates, parameters, outputs, change sets, and drift detection.



Review the purpose of `template.yaml`, functions, APIs, and permissions.

Lab 6 - Steps 6-10



Open Elastic Beanstalk.



Open Amazon ECR.



Open ECS.



Review environments, application versions, platforms, and deployment policies.



Review container image repository concepts.

Lab 6 - Steps 11-15



Review clusters, task definitions, services, and tasks.



Design a deployment path for a small serverless API.



Save your notes.



Create a comparison table for CloudFormation, SAM, Elastic Beanstalk, ECS, and Lambda console deployment.



Design a deployment path for a containerized web application.

Lab 6 - Verify and Clean Up

Verification

1. Deployment service comparison table
2. Serverless deployment design
3. Container deployment design
4. CloudFormation and SAM notes

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep rg-az900-labs only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 16 performs the final course cleanup.

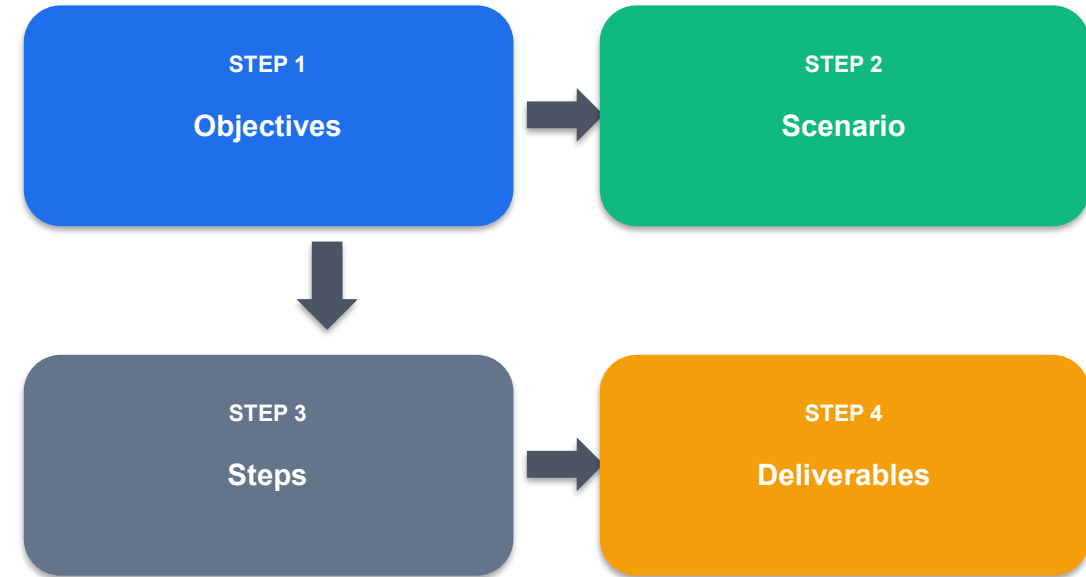
Lab 7 - Application Security, Secrets, Cognito, KMS, and Least Privilege

Domain 3 - Troubleshooting and Optimization

Day 2

labs/lab-07-security-
secrets-cognito-
kms.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 7 - Objectives

- Secure application secrets and configuration.
- Review Cognito authentication and authorization patterns.
- Understand KMS encryption for application data.
- Apply least-privilege and resource-based permissions.

Lab 7 - Steps 1-5



Open Secrets Manager.



Open Systems Manager Parameter Store.



Open Amazon Cognito.



Review secret storage, rotation, versions, and access policies.



Compare standard parameters, secure strings, and hierarchy naming.

Lab 7 - Steps 6-10



Review user pools, identity pools, app clients, and hosted UI concepts.



Review AWS managed keys, customer managed keys, key policies, and grants.



Review IAM identity-based policies and resource-based policies.



Open KMS.



Review Lambda environment variable encryption.

Lab 7 - Steps 11-14

- Compare API Gateway IAM authorization, Lambda authorizers, and Cognito authorizers.
- Create a secure application checklist.
- Map where secrets, config, tokens, encryption keys, and permissions belong in an application.
- Save your notes.

Lab 7 - Verify and Clean Up

Verification

1. Secrets Manager vs Parameter Store comparison
2. Cognito concept notes
3. KMS encryption notes
4. Secure application checklist

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep rg-az900-labs only when the next lab needs it.
3. Confirm no paid resource is left running.
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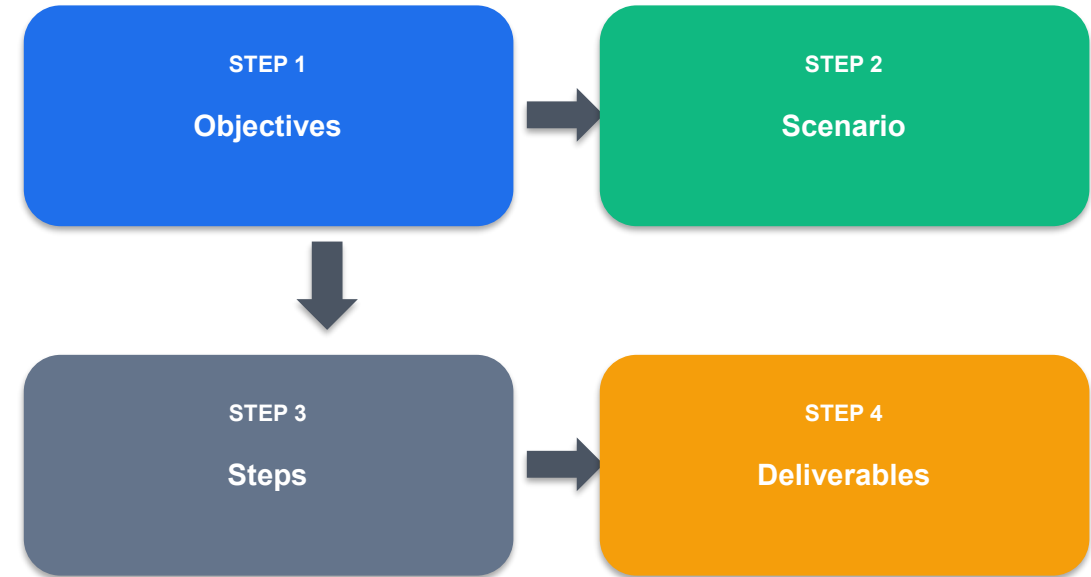
Lab 8 - Monitoring, Troubleshooting, Optimization, and Exam Review

Domain 3 - Troubleshooting and Optimization

Day 2

labs/lab-08-
monitoring-
troubleshooting-
optimization-exam-
review.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 8 - Objectives

-  Review application observability tools.
-  Diagnose common Lambda, API Gateway, DynamoDB, and deployment issues.
-  Identify performance and cost optimization patterns.
-  Complete certification-style review practice.

Lab 8 - Steps 1-5



Open CloudWatch.



Review Lambda logs and metrics such as errors, duration, throttles, and concurrent executions.



Open X-Ray.



Review metrics, logs, log groups, dashboards, and alarms.



Review API Gateway metrics and access logs conceptually.

Lab 8 - Steps 6-10



Review traces, segments, service maps, and sampling concepts.



Create a troubleshooting checklist for Lambda permission errors, timeout errors, throttling, API errors, and DynamoDB throttling.



Review DAX, ElastiCache, CloudFront, and API Gateway caching use cases.



Open CloudTrail and review audit events.



Create an optimization checklist covering memory tuning, retries with backoff, caching, connection reuse, SDK pagination, and reserved concurrency.

Lab 8 - Steps 11-15



Map each lab to exam domains.



Mark weak topics by domain.



Save final notes.



Complete a timed 30-question practice set from trainer-provided questions or AWS practice resources.



Clean up temporary resources if instructed.

Lab 8 - Verify and Clean Up

Verification

1. Monitoring and troubleshooting checklist
2. Optimization checklist
3. Exam-domain revision map
4. Timed practice score

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep rg-az900-labs only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 16 performs the final course cleanup.

What You Achieved

Develop

**AWS
services**

Built lab-aligned patterns

Secure

**Least
privilege**

Protected credentials and
data

Deploy

Automation

Compared CI/CD and IaC

Operate

**Observabilit
y**

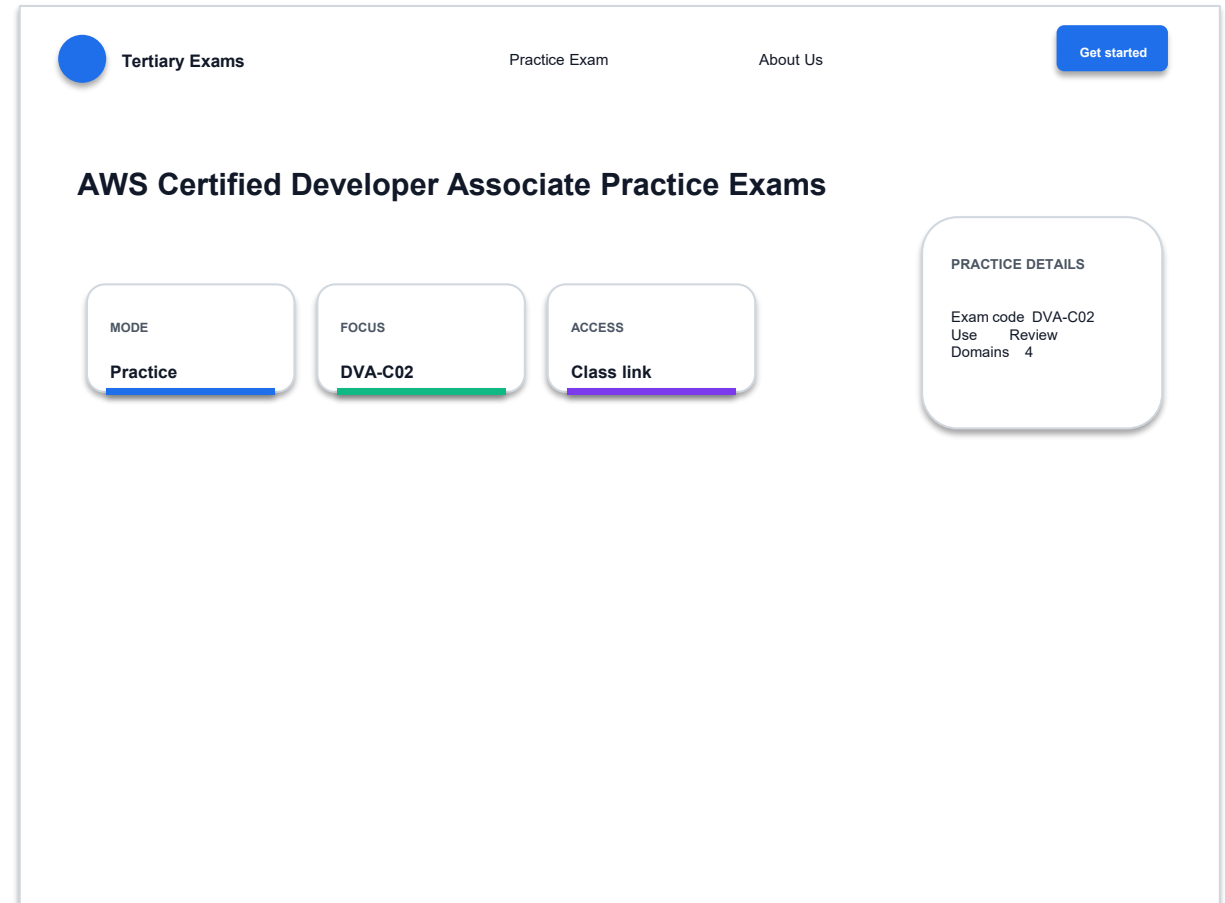
Diagnosed and optimized

Take the Online AWS Developer Practice Exams

- Reinforce AWS Developer Associate domains with realistic, exam-style questions before assessment day.
- Test development, security, deployment, troubleshooting and optimization decisions.
- Review mode helps you check the correct answer and revisit the concept after each question.
- Complete the practice questions from Google Classroom, then bring difficult items to review.

Access and practise at:

[Google Classroom > Practice Exams](#)



Courseware and Assessment on the LMS

 Download courseware from <https://lms-tms.tertiaryinfotech.com/>

 Complete the WA and PP on the LMS.

 Keep your learner evidence ready for assessment.

REMINDER

Assessment

 Day 2, 4:00 PM-6:00 PM

 WA: 1 hour | PP: 1 hour

 Open book

 Take assessment attendance and submit on LMS

Assessment Flow Reminder



The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

Digital Attendance (Mandatory)

 Complete TRAQOM - SSG digital attendance.

 Use the LMS QR code provided by the trainer.

 Confirm successful submission before leaving.

CLOSE

Thank You



enquiry@tertiaryinfotech.com



www.tertiarycourses.com.sg